

Professional Summary

Mechatronics Systems undergraduate supporting semiconductor automation with technicians and vendors. Developed and executed 50+ test cases, FMEA documentation, and root-cause analysis while delivering production go-lives, equipment recovery, data automation, technical work instructions, and time-critical repairs.

Technical Skills

Automation & controls	AMHS/ASRS, AMR/AGV, DPL, strapping/unstrapping, handler systems, Fleet Manager, e-WareNavi, system integration, HMI testing, safety isolation
Data & software	Python, PowerShell, JavaScript, C/C++, SQLite, REST APIs, Tableau, Excel, Outlook automation, Microsoft 365 Copilot, OpenCV
Embedded & design	SolidWorks, CAD, RTOS, Arduino, ARM Assembly, SPI, I2C, GPIO
Testing & delivery	UAT, FMEA, RCA, Cpk, 2-sample t-tests, variance analysis, hardness/case-depth testing, supplier/vendor coordination, technical documentation

Experience

Automation Engineer Intern (IWSP) | Infineon Technologies Asia Pacific Pte LtdSep 2025 - Aug 2026 | Singapore

- Partnered with technicians and vendors to recover AMHS, ASRS, AMR/AGV, OHT/OHV, DPL, strapping, and handler systems using Fleet Manager, e-WareNavi, and material-control applications; performed preventive maintenance and used a multimeter and hand tools to replace and realign a worn stopper within a 30-minute window, then verified all six drop-point levels.
- Designed and executed 50+ OHV UAT cases plus stocker, feeder-table, and Lights Out 2.0 integration/exception tests; prepared FMEA documentation, coordinated vendor fixes, and verified production go-live, including feeder automation that replaced manual Traystack loading.
- Performed system/server recovery for storage, transportation, and remote-command platforms; supported software rollouts and periodic OS/antivirus patching; resolved basic computer hardware/software issues independently or with IT engineers.
- Conducted RCA on DPL barcode misalignment and coordinated supplier corrective actions; built a feedback-driven Python issue tracker, a PowerShell/Outlook low-stock alert for 36 Priority 1 items, and seven live Tableau OHV views tracking alarms and uptime against a 99% target.
- Created three controlled 12-hour OJTIs from scratch for OHV, stocker, and feeder equipment covering safety, operation, alarm recovery, and competency checks; trained operators and tracked technician certification and refresher training.

Signals | Singapore ArmyMay 2021 - May 2023
Completed full-time National Service in a Signals unit.

Process Development Engineer Intern | Shimano (Singapore) Pte LtdMar 2020 - Aug 2020
Validated a new steel source for heat-treated gears using 30 samples across 10 tray locations; performed Rockwell/Vickers hardness, 0.6 mm case-depth and metallographic testing, then Cpk, 2-sample t-test and variance analysis to confirm specifications with no process changes required.

Education

Bachelor of Engineering (Honours), Mechatronics SystemsExpected Apr 2027
Singapore Institute of Technology

Diploma in Mechatronics and Robotics EngineeringApr 2021
Singapore Polytechnic

Selected Projects

- Autonomous Trolley Retrieval System** | 2023 - 2025: Designed a custom gripper and integrated OpenCV trolley detection, navigation, obstacle avoidance, and handle sanitisation on an autonomous mobile robot.
- Privacy-First Job Apply Assistant** | 2026: Built a Python/SQLite assistant that aggregates job APIs, ranks roles against PDF/DOCX resume skills, tracks applications, and provides safe Manifest V3 autofill without submission.
- Multi-provider AI Chrome Assistant** | 2026: Developed a privacy-focused Chrome side-panel assistant supporting eight AI providers, selectable models, webpage-context analysis, and local API-key handling; packaged a secret-free repository release.

Certifications & Recognition

Certified SOLIDWORKS CAD Design Associate (2025) • Google Project Management Professional Certificate (2025) • Cisco CCNA (issued 2022) • bizSAFE Level 2 (2020)